

Electric Vehicle Population Data Analysis

Assignment for Internship – GrowAI

Submitted By: Ishu Verma

Date: 03/06/2026

Introduction

This report presents a detailed analysis of the Electric Vehicle Population Dataset provided by the Washington State Department of Licensing. The dataset contains information regarding Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) registered across Washington State.

The purpose of this analysis is to clean and prepare the data, explore adoption patterns and vehicle characteristics, visualise important trends, and build a predictive model for estimating electric vehicle range.

The analysis was performed using Python libraries including Pandas, NumPy, Matplotlib, Seaborn, Plotly, Folium, and Scikit-learn.

The main objectives of this project include:

- Cleaning and preparing the dataset
- Identifying missing values and duplicate records
- Protecting privacy through VIN anonymisation
- Exploring trends in EV adoption and registration
- Visualising county-wise and geospatial distribution
- Building and evaluating a Linear Regression model for electric range prediction

Section 1: Data Cleaning

1.1 Missing Values Analysis

The dataset was examined to identify missing values and determine which columns required cleaning.

The following columns contained missing values:

Column	Missing Values
Legislative District	733
Vehicle Location	18
County	10
City	10
Electric Utility	10
Postal Code	10
2020 Census Tract	10
Electric Range	8

The analysis shows that most columns contain minimal missing information, while Legislative District has the highest number of missing records.

1.2 Handling Base MSRP and Electric Range

The assignment required analysis of Base MSRP and Electric Range.

Base MSRP

The dataset does not contain a Base MSRP column. Therefore, no cleaning or imputation process was applied to this variable.

Electric Range

The Electric Range column contained 8 missing values.

To preserve dataset completeness and avoid bias from extreme values, missing values were replaced using the median electric range.

The dataset also contained 184,285 zero values in Electric Range.

These zero values were retained because they represent valid observations, particularly among Plug-in Hybrid Electric Vehicles where electric-only range may be minimal or unavailable.

1.3 Duplicate Records

Duplicate observations were checked using Python duplicate detection functions.

The analysis showed:

Total duplicate records: 0

No duplicate entries were identified, so no rows were removed.

1.4 VIN Anonymisation

Vehicle Identification Numbers (VINs) contain sensitive information and therefore required anonymisation.

To maintain privacy while preserving uniqueness, each VIN was converted into a secure SHA256 hash value.

The process involved:

- Creating a new column called hashed_vin
- Transforming VIN values using SHA256 hashing
- Removing the original VIN column

This approach ensures privacy protection without affecting analytical accuracy.

1.5 Cleaning Vehicle Location

The vehicle_location column contained GPS coordinates stored in text format such as:

POINT (-122.3321 47.6062)

These coordinates were parsed and separated into two numerical columns:

- Longitude
- Latitude

This transformation improved readability and prepared the dataset for geospatial analysis and mapping.

Section 2: Exploratory Data Analysis

2.1 Top 5 EV Makes and Models

The most frequently registered EV manufacturers are:

Make	Registrations
TESLA	117,384
CHEVROLET	19,675
NISSAN	16,121
FORD	15,637
KIA	14,328

The most frequently registered EV models are:

Model	Registrations
MODEL Y	62,150
MODEL 3	38,314
LEAF	13,478
MODEL S	7,815
BOLT EV	7,692

Interpretation

Tesla dominates the EV market, both at manufacturer and model levels. Model Y and Model 3 account for a significant proportion of total registrations.

2.2 County Distribution

County-wise registration analysis revealed substantial regional variation.

King County recorded the highest EV registrations with:

140,181 vehicles

Interpretation

EV adoption is concentrated in urban and economically developed counties where charging infrastructure and consumer awareness are stronger.

2.3 EV Adoption by Model Year

Vehicle registrations were analysed according to model year.

The data indicates:

- Slow growth during early years
- Gradual expansion after 2010
- Rapid acceleration beginning around 2018
- Peak adoption around 2023

Interpretation

This trend reflects increasing market maturity, improved battery technology, and greater public acceptance of electric mobility.

2.4 Average Electric Range

The average electric range across all vehicles is: **38.15 miles**

Interpretation

This relatively low value is influenced by the large proportion of Plug-in Hybrid Electric Vehicles with limited or zero electric-only range.

2.5 CAFV Eligibility

Clean Alternative Fuel Vehicle eligibility results are:

Status	Percentage
Eligibility Unknown	64.48%
CAFV Eligible	27.12%
Not Eligible	8.41%

Interpretation

Most vehicles fall into the “Eligibility Unknown” category, largely due to PHEV classifications and incomplete battery research data.

2.6 Electric Range Across Makes and Models

Electric range varies substantially across manufacturers and vehicle models.

High-range makes include:

- Jaguar
- Wheego Electric Cars
- THINK

High-range models include:

- Roadster
- Bolt EV
- I-PACE

Interpretation

Vehicle range depends strongly on battery capacity, drivetrain technology, and vehicle design.

2.7 Base MSRP Analysis

Average Base MSRP analysis could not be conducted because the dataset does not include a Base MSRP field.

2.8 Regional Trends

The analysis identified clear urban-rural differences.

Major urban areas including:

- Seattle
- Bellevue
- King County

showed significantly higher EV adoption compared with rural counties.

Interpretation

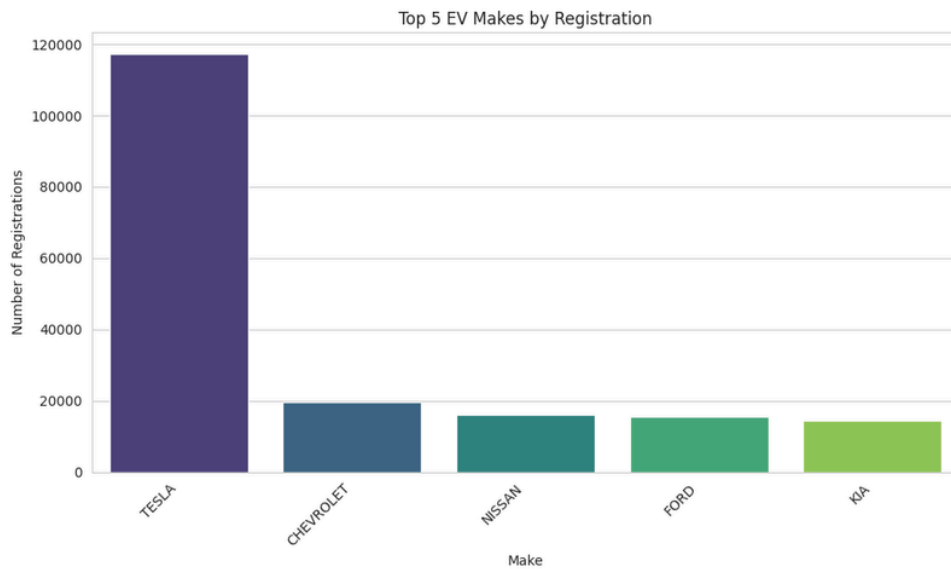
Infrastructure availability, population density, and economic conditions appear to be major drivers of EV adoption.

Section 3: Data Visualisation

The following visualisations were developed to support the analysis.

3.1 Bar Chart – Top EV Makes and Models

The bar chart highlights dominant manufacturers and vehicle models.

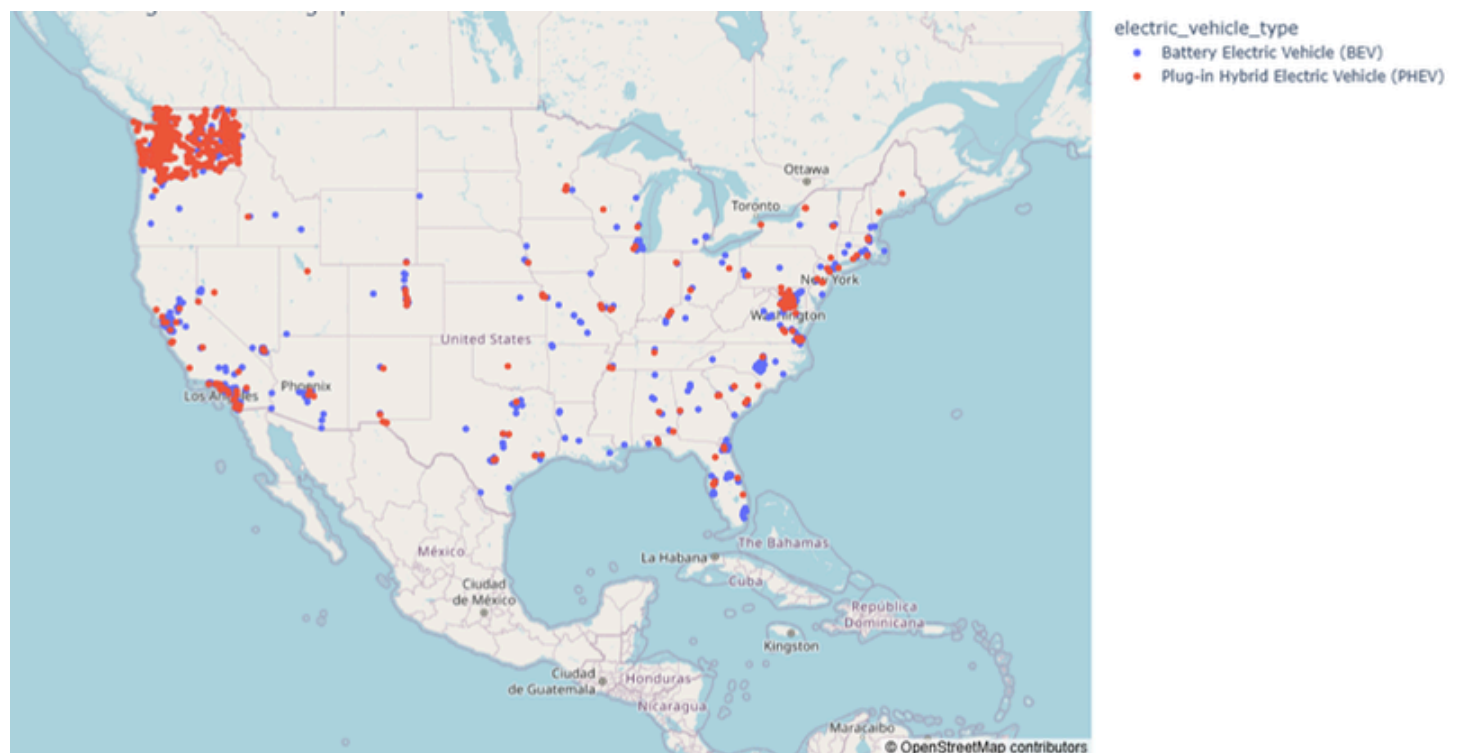


Interpretation

Tesla significantly outperforms competing brands.

3.2 County Distribution Map

A geospatial map was created to visualise county-wise EV concentration.

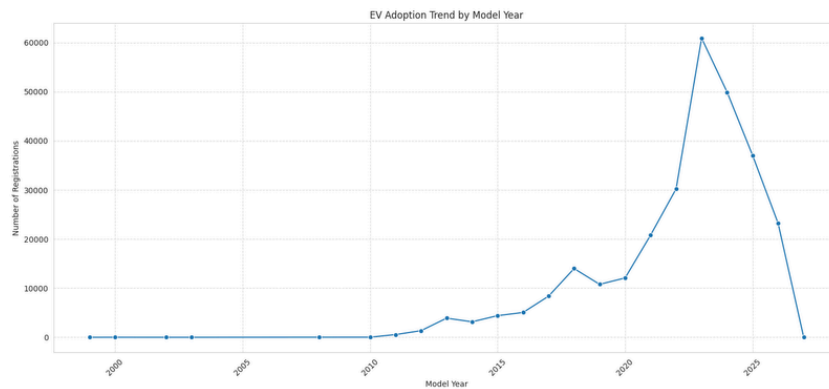


Interpretation

The highest density occurs near Seattle and surrounding metropolitan areas.

3.3 Model Year Adoption Trend

A line chart illustrates EV adoption over time.

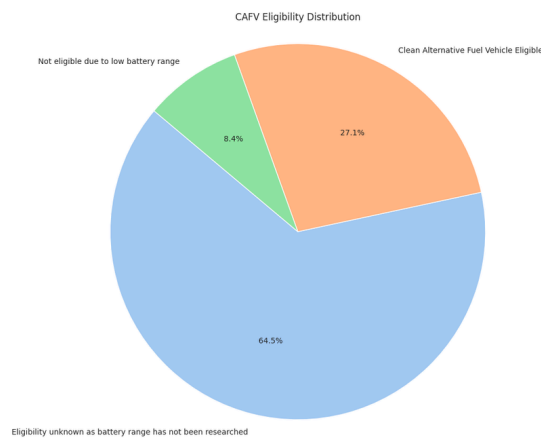


Interpretation

The steep increase after 2018 confirms strong market expansion.

3.4 CAFV Eligibility Pie Chart

A pie chart illustrates eligibility distribution.



Interpretation

Only a minority of vehicles are clearly CAFV-eligible.

3.5 Geospatial Vehicle Location Map

A Folium map with marker clusters was created using latitude and longitude coordinates.

Interpretation

Vehicle registrations are concentrated in urban areas with established charging networks.

Section 4: Linear Regression Model

4.1 Predicting Electric Range

A Linear Regression model was developed to predict vehicle electric range.

The selected features included:

- Model Year
- Make
- Model
- Electric Vehicle Type

4.2 Handling Categorical Variables

Categorical features were transformed using:

One-Hot Encoding

using `pd.get_dummies(drop_first=True)`.

This converted text variables into numerical binary indicators suitable for regression analysis.

4.3 Model Performance

The Linear Regression model achieved:

Metric	Value
R ² Score	0.51
MAE	36.86
RMSE	54.30

Interpretation

The model explains approximately 51% of variation in electric range.

While useful, additional variables are needed for stronger predictive accuracy.

4.4 Influence of Features

The model showed:

- Model Year coefficient: -16.33
- Strong influence from specific makes and models

Interpretation

The negative coefficient for model year may reflect growing representation of lower-range PHEVs in newer registrations.

Base MSRP influence could not be evaluated because the dataset lacked this feature.

4.5 Improving Model Accuracy

Future improvements may include:

- Battery capacity information
- Vehicle weight
- MSRP data
- Aerodynamic features
- Separate modelling for BEVs and PHEVs
- Random Forest or XGBoost models
- Regularisation techniques such as Ridge and Lasso

These improvements could substantially increase predictive performance.

4.6 Predicting New EV Models

The regression model can estimate electric range for new vehicles provided the same input features are available.

However, caution is required because:

- Predictions may contain errors
- New unseen models may reduce accuracy
- Negative predictions may require adjustment

Therefore, predictions should be interpreted alongside model error metrics.

Conclusion

This analysis produced several important findings regarding electric vehicle adoption in Washington State.

Key conclusions include:

1. Tesla dominates the EV market, particularly through Model Y and Model 3.
2. King County represents the centre of EV adoption.
3. EV registrations increased dramatically after 2018.
4. Average electric range appears low because many vehicles are PHEVs.
5. Urban regions display much higher adoption than rural areas.
6. Only about 27% of vehicles are clearly CAFV-eligible.
7. Linear Regression explains approximately 51% of electric range variation.

The absence of Base MSRP limited pricing analysis and reduced model capability.

Future work should integrate additional technical and pricing variables to develop more accurate prediction models.

Appendix

Tools Used

- Python 3
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- Folium
- Scikit-learn

Data Source : Electric Vehicle Population Data – data.gov

Repository Link : <https://github.com/ishhverma/EV-Data-Analysis>

For full code implementation and visualisations, refer to the accompanying Jupyter Notebook:

EV_Analysis_Ishu_Verma.ipynb